

Cost Analysis

2026-09-01

Contents

1	Assumptions	1
2	Executive Summary	1
3	Breakdown by grades	8

1 Assumptions

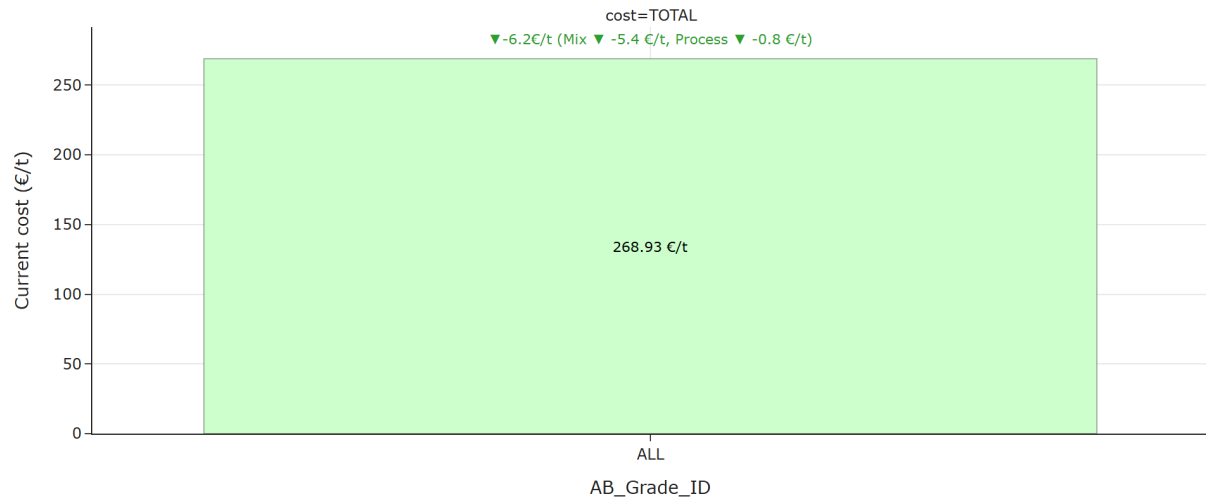
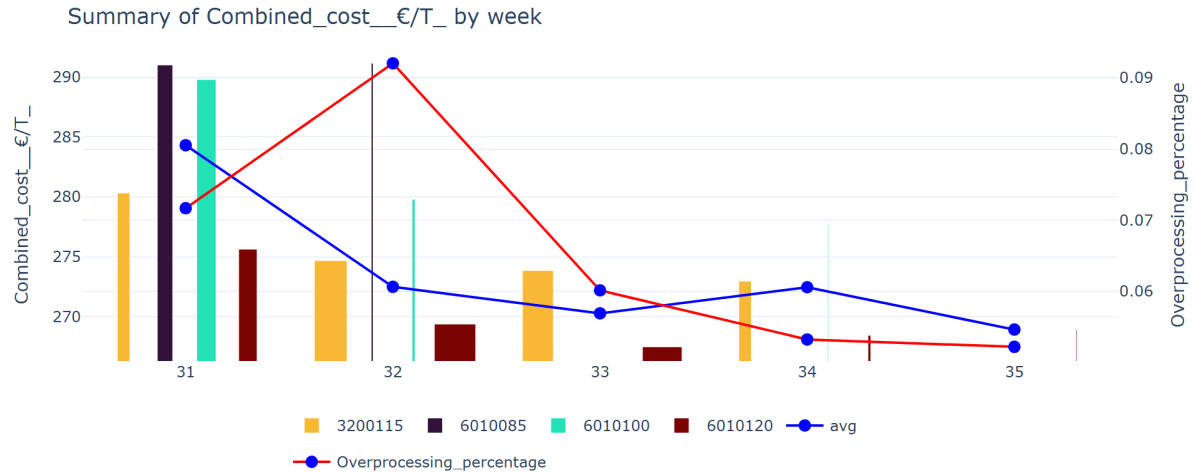
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

- **Period Analyzed:** July 26, 2026 – August 31, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
 - **Current:** Refers to the last week of the analyzed period.
 - **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
 - **Best:** Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

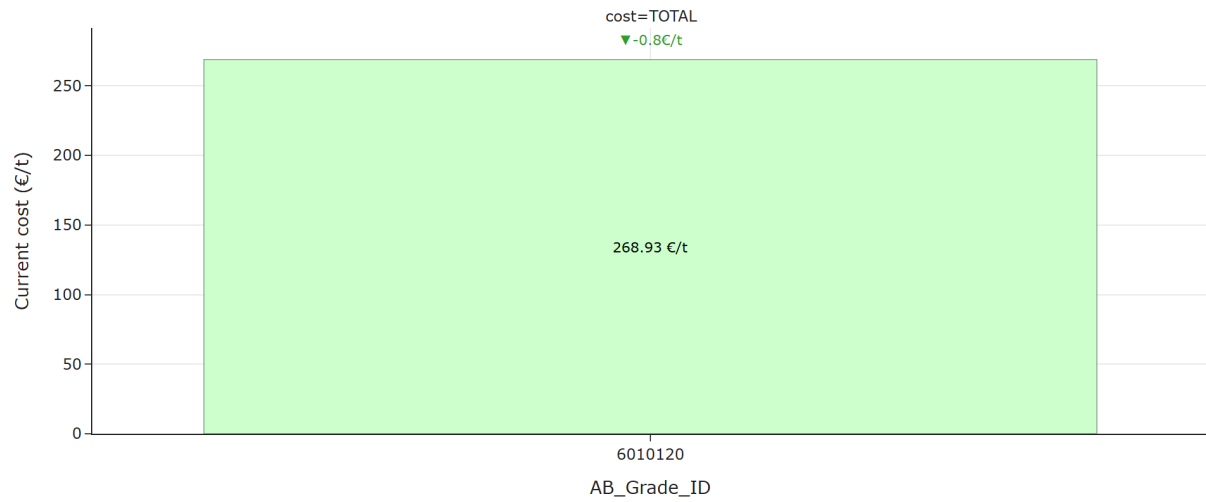
The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw a decrease of 6.24 €/t, resulting in 268.93 €/t.

Breaking down the overall cost change, -5.42 €/t was driven by a cheaper mix of grades, and -0.82 €/t was due to an improvement in the process.



For grade 6010120, TOTAL cost saw a decrease of 0.82 €/t, resulting in 268.93 €/t.



Average change by component:

- **Electricity_€/*T*** decreased on average (-1.31 €/t).
- **Steam_€/*T*** increased on average (+1.06 €/t).
- **Chemicals_€/*T*** decreased on average (-0.35 €/t).
- **Fibre_cost_€/*T*** decreased on average (-0.14 €/t).
- **Starch_€/*T*** decreased on average (-0.07 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_€/*T*** (+1.06 €/t on average), with the highest increase in grade 6010120 (+1.06 €/t) and the largest average decrease comes from **Electricity_€/*T*** (-1.31 €/t on average), with the biggest drop in grade 6010120 (-1.31 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Predryers** increased on average (+2.39 €/t).
- **Steam_Afterdryers** decreased on average (-2.08 €/t).
- **Steam_Whitewater** increased on average (+0.71 €/t).
- **Steam_Starchkitchen** increased on average (+0.09 €/t).
- **Steam_Heatexchangers** decreased on average (-0.05 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Predryers** (+2.39 €/t on average), with the highest increase in grade 6010120 (+2.39 €/t) and the largest average decrease comes from **Steam_Afterdryers** (-2.08 €/t on average), with the biggest drop in grade 6010120 (-2.08 €/t). These components stand out and may require further investigation.

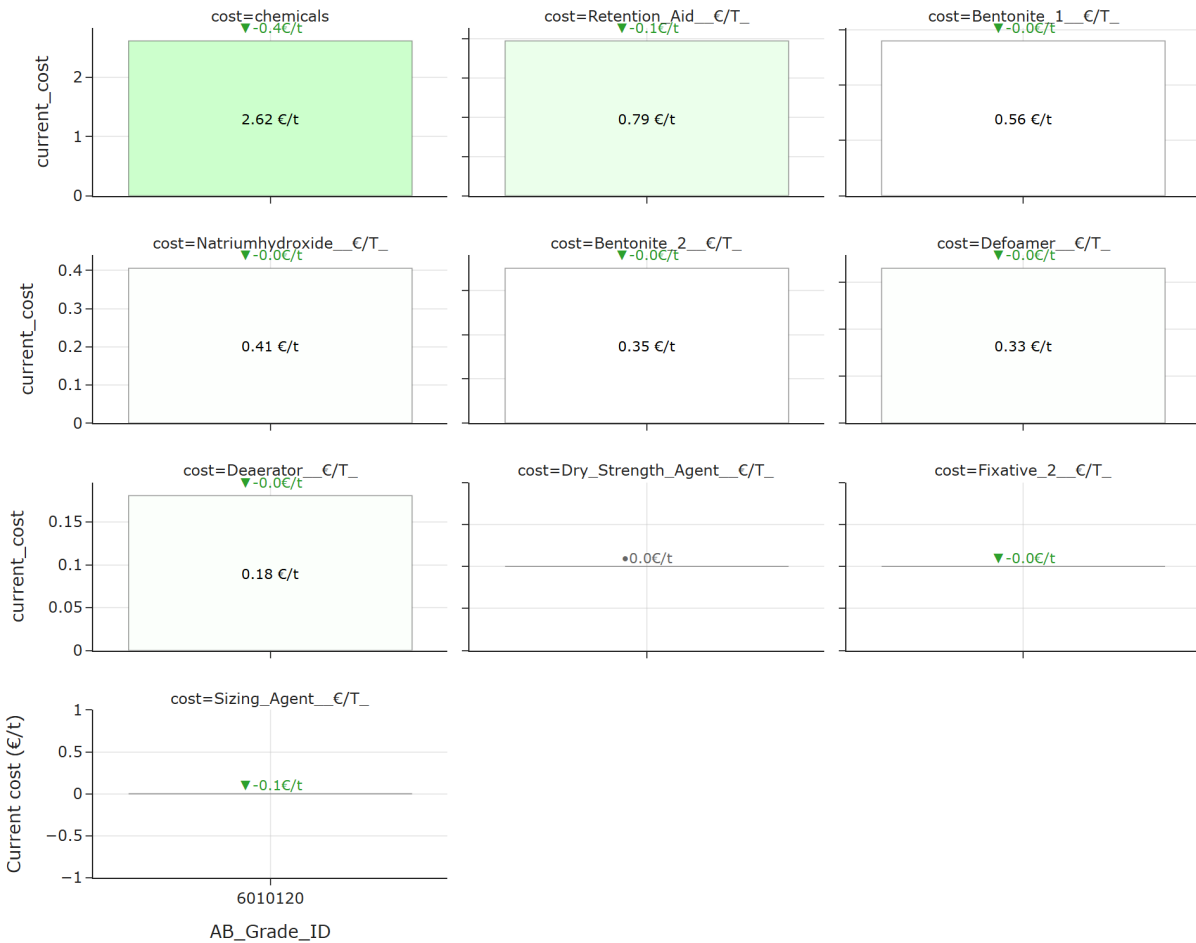


Average change by electricity:

- **Electricity_Specific** decreased on average (-1.60 €/t).
- **Electricity_Freqconverters** increased on average (+0.32 €/t).

- **Electricity_Other** decreased on average (-0.24 €/t).
- **Electricity_Predryer** increased on average (+0.17 €/t).
- **Electricity_Press_and_Steambox** increased on average (+0.11 €/t).
- **Electricity_Wiresection** decreased on average (-0.10 €/t).
- **Electricity_Afterdryer** increased on average (+0.03 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Freqconverters** (+0.32 €/t on average), with the highest increase in grade 6010120 (+0.32 €/t) and the largest average decrease comes from **Electricity_Specific** (-1.60 €/t on average), with the biggest drop in grade 6010120 (-1.60 €/t). These components stand out and may require further investigation.

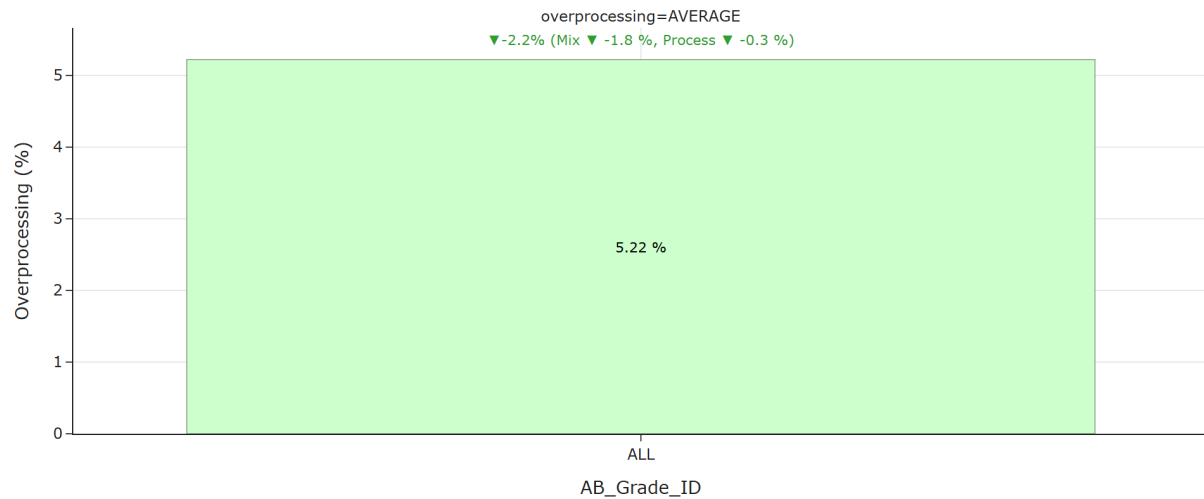


Average change by chemicals:

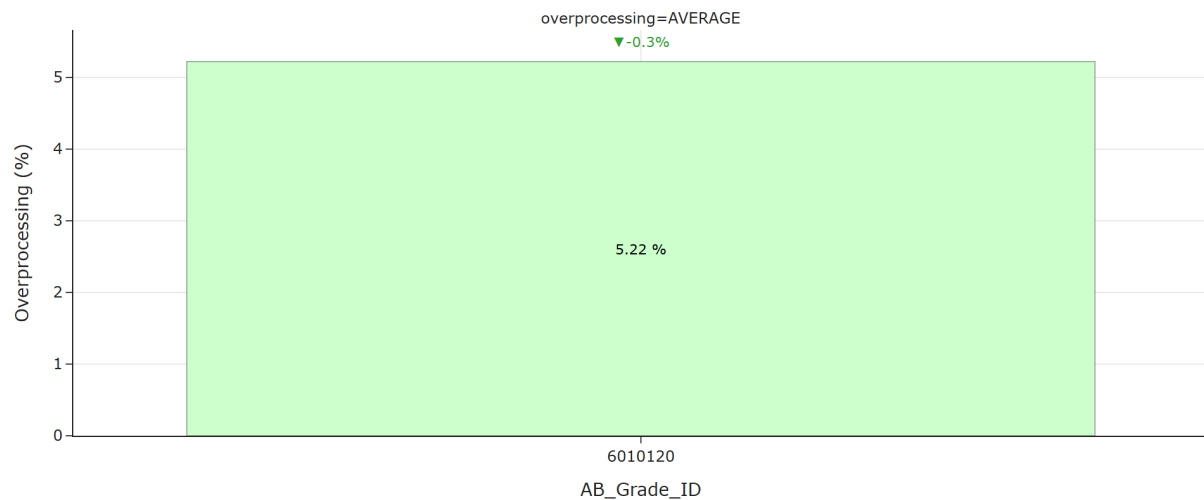
- **Retention_Aid_€/T** decreased on average (-0.14 €/t).
- **Deaerator_€/T** decreased on average (-0.03 €/t).

- **Defoamer_€/*T*** decreased on average (-0.02 €/t).
- **Natriumhydroxide_€/*T*** decreased on average (-0.02 €/t).
- **Bentonite_1_€/*T*** decreased on average (-0.00 €/t).
- **Bentonite_2_€/*T*** decreased on average (-0.00 €/t).

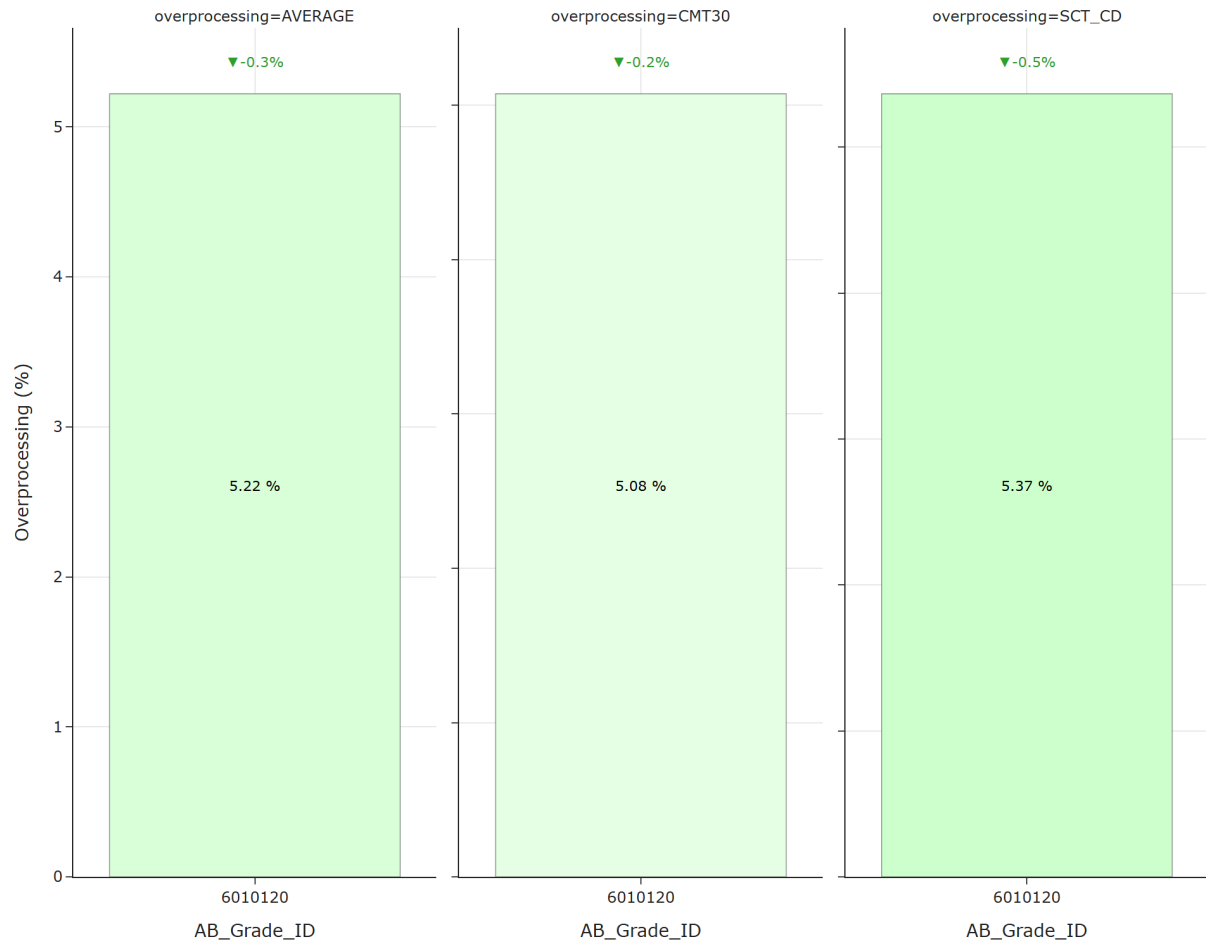
A more detailed breakdown by component shows that the largest average decrease comes from **Retention_Aid_€/*T*** (-0.14 €/t on average), with the biggest drop in grade 6010120 (-0.14 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw a decrease of 2.16 %, resulting in 5.22 %.



For grade 6010120, TOTAL overprocessing saw a decrease of 0.34 %, resulting in 5.22 %.



Average change by component:

- **SCT_CD** decreased on average (-0.45 %).
- **CMT30** decreased on average (-0.23 %).

A more detailed breakdown by component shows that the largest average decrease comes from **SCT_CD** (-0.45 % on average), with the biggest drop in grade 6010120 (-0.45 %). These components stand out and may require further investigation.

3 Breakdown by grades